

Evidence-Carrying Live Search: Budgeted Multimodal Retrieval for Interactive Video Browsing

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Abstract. Interactive video retrieval often fails at the handoff from a ranked hit to submit-ready evidence: a relevant frame can lose its media identity, become impossible to reopen, or trigger an expensive verification path too late. We present TGLTW-RMIT, a live-first multimodal system built around an evidence-carrying candidate contract and a budgeted precision ladder. Offline processing gives selected frames and shot parents stable identifiers, joins them with timestamps, OCR, speech, metadata, and VLM fields, and stores parent-to-frame links. The web path fuses dense and payload-text retrieval with reciprocal-rank fusion, temporal coherence, scene diversity, and bounded reranking; exact search, larger HNSW pools, verification, and in-video search are request-scoped escalations. KIS-V, KIS-T, KIS-C, AVS, and VQA use different policies over the same evidence identity. Grounded VQA fails closed on unavailable media, malformed output, invalid confidence, or missing evidence. We give implementation details and a frozen live/offline evaluation protocol; no leaderboard numbers are claimed before the corpus, index, and runtime are frozen. Source and deployment links are on the public [project page](#).

Keywords: Video retrieval · Multimodal search · Interactive retrieval · Video Browser Showdown · Grounded VQA

1 Introduction

The Video Browser Showdown evaluates exploratory video search in a timed public setting: an operator interprets a task, retrieves evidence, inspects media, and submits a temporal result before the clock expires. The 2027 call emphasizes KIS-T, KIS-C, VQA, and AVS, with KIS-V as a minority task, and asks for a six-page demo paper with a screenshot and interactive-search description [1]. Earlier VBS work also treats interaction traces and human-facing search as evaluation objects [2,3]. Predictable latency is therefore a system property, not only an implementation detail.

TGLTW-RMIT keeps a fast path over existing indexes: approximate HNSW retrieval [4], payload search, rank fusion, and bounded reranking. Exact search, larger pools, verification, and in-video refinement are explicit per-request actions. The retrieval unit is consequently a reopenable evidence object: relevance is insufficient if media identity is lost, evidence cannot be loaded again, or an answer cannot be submitted safely.

The novelty is system-level rather than a new embedding model. The implementation contributions are:

1. **Evidence-carrying indexing:** deterministic frame/shot identifiers, parent links, temporal fields, and inspectable multimodal payloads survive rank fusion;
2. **A budgeted precision ladder:** dense/payload retrieval, fusion, coherence, and diversity form the default, while exact search, larger pools, verification, and in-video search expose measurable cost;
3. **A task and grounding contract:** five VBS modes reuse identity but select different interaction policies, and web VQA refuses an answer when evidence or the response is invalid.

HNSW, RRF, VLM reranking, and grounded VQA are not claimed as individually new. Contract tests cover the implementation, while accuracy and latency remain pending a frozen query set, index manifest, and runtime configuration. Source, paper, and deployment links are on the public [project page](#).

2 VBS Setting and System-Level Delta

VBS systems combine semantic representations with fast browsing, feedback, and interaction-aware evaluation [2,5,3,6]. Local challenge systems provide precedents for OCR/ASR evidence, query expansion, temporal search, reranking, and feedback [7,8]. TGLTW-RMIT uses Qdrant-style collections, dense/payload retrieval, reciprocal-rank fusion (RRF) [9], and bounded operator actions; its rewriting and feedback remain narrower than open-ended expansion or reweighting [10,11].

System-level delta. Three goals drive the design: keep the common path fast, attribute expensive precision actions to the operator, and preserve evidence identity from retrieval through submission. Relative to the legacy running snapshot, the instrumented path makes provenance an invariant, task routing and escalation bounded policies, and VQA evidence tied to the media shown. The claims are falsifiable through identity parity, the quality–latency frontier of each escalation, and grounded-answer/failure rates; legacy records remain source/timestamp-level until migration is audited.

The planned corpus combines V3C, marine video, and GynSurg/LapGynLHE collections, with V3C dominant for most tasks [1,12,13,14]. The final run must freeze accessible shards, video IDs, checksums, keyframe/shot version, schema, and data agreements; legacy and source-index states are reported separately until re-indexing is audited.

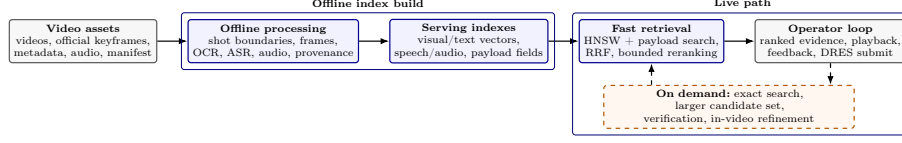


Fig. 1. TGLTW-RMIT’s evidence-carrying, live-first control plane on the instrumented path. The offline index supplies a fast common path, while exact search, verification, and in-video refinement are budgeted escalations rather than mandatory steps for every query.

3 TGLTW-RMIT Architecture

3.1 Offline evidence and provenance

Preprocessing prefers official V3C shot maps, keyframes, metadata, and ASR; otherwise it uses TransNetV2 [15]. A compressed-size prefilter, CLIP variance [16], farthest-point sampling, and sharpness bonus set a bounded frame budget. OCR, objects, faster-whisper speech [17], VLM narratives, and optional audio/visual vectors enrich the payload; speech is joined to nearby visual text, while ambient audio remains a separate collection.

The central structure is a two-level evidence graph. Each frame has a deterministic point ID and payload (media, native frame index/timestamp, shot, captions, OCR, objects, events, speech, and provenance). Each shot has a normalized aggregate and representative frame IDs in `vbs_shot_index`; coarse-to-fine routing stays disabled until recall is measured. Timestamp/ordinal fallbacks support legacy records, while web VQA prefers native indices. A candidate is therefore a reopenable evidence object shared by browser, model input, log, and submission adapter.

3.2 Retrieval and interaction

Text queries use bounded CQR and at most one HyDE description. The web path fuses original, HyDE, optional secondary-vector, and payload full-text lists with $\text{RRF}(x) = \sum_{\ell \in \mathcal{L}} (k + \text{rank}_{\ell}(x))^{-1}$ [9]. The lexical branch is a full-text gate with rank-position, not calibrated BM25, scores. Temporal coherence, scene diversity, `matched_via` labels, playback, feedback, in-video search, and DRES submission complete the browser path; JSONL interaction events are written locally before best-effort DRES logging.

Budgeted precision ladder. The default uses approximate dense search and a small reranking head. Request fields can select exact search, larger HNSW effort, verification, or in-video search after a promising video is identified. H-EAGLE-lite, secondary SigLIP, and verification remain opt-in implemented branches, not measured defaults.

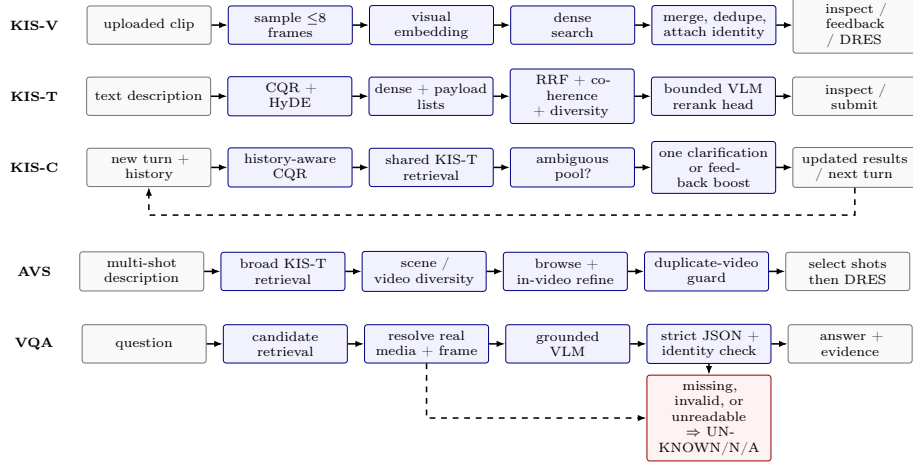


Fig. 2. Compact task flowcharts. KIS-V bypasses text expansion; KIS-T is the shared text route; KIS-C adds bounded state and clarification; AVS adds cross-video coverage and a duplicate guard; VQA adds candidate-specific media grounding and fails closed.

4 Task-Specific Live Pipelines

The five VBS routes share candidate identity, evidence, feedback, and submission primitives but choose different entry and operator policies. KIS-T/KIS-C/AVS use the Type-1 web backend, VQA the Type-2 backend, and KIS-V dedicated image/video endpoints; the inherited CLI Type-3 experiment is not an additional 2027 task. Figure 2 summarizes the routes.

KIS-V. A temporary uploaded clip is sampled at most eight readable positions. Each sample is dense-searched and merged by indexed point, preserving the strongest hit and query timestamp. The endpoint bypasses text expansion, payload-text matching, and VLM reranking; cards retain source video, frame, timestamp, and submission actions.

KIS-T. Text enters Type-1 CQR/HyDE, then original, optional secondary, and payload lists are fused with RRF, coherence, and diversity. A small VLM head reranks the leading pool; exact search, larger HNSW, and verification are explicit overrides.

KIS-C. KIS-C is stateful policy around KIS-T. Ambiguity averages distinct-video ratio and normalized top-1/top-2 margin; threshold 0.7 gates one facet-specific clarification. The answer boosts its candidate pool, while Rocchio feedback and confirmed/rejected descriptions update the next CQR prompt. No autonomous planner is used.

AVS. AVS reuses broad KIS-T retrieval but optimizes cross-video coverage. Coherence/diversity reduce near duplicates; the operator groups, browses, and selects shots. A per-task set gives a soft duplicate warning, while DRES remains authoritative.

VQA. VQA is an evidence-integrity layer. A candidate reaches the VLM only after a real keyframe/video frame is resolved inside the dataset root, preferring native frame index over timestamp. The parser requires `found=true`, nonempty answer, and confidence in $[0, 1]$; missing media, invalid JSON, timeout, unknown answer, or invalid confidence returns UNKNOWN/N/A with zero score and no identity. Valid answers retain candidate, video, and evidence frame/timestamp. This fail-closed contract is scoped to the web route; CLI VQA remains a migration target.

SCREENSHOT NEEDED

Composite live UI capture: left: task/query/results; right: VQA grounding/evidence; optional AVS/DRES crop

Fig. 3: Composite live UI capture. Replace before submission.

5 Evaluation Protocol

The paper separates the official live DRES outcome from controlled offline replay. Before competition, the REST client and JSONL log must be checked against the exact VBS OpenAPI contract; the hand-written wrapper is currently unverified. The live record stores task/session IDs, mode, timings, queries/clarifications, candidates, submissions, judge outcome, and model/index/configuration hashes. Reports retain score/rank, time-to-first/correct submission, submission counts, interaction trace, KIS-C clarification, AVS ranges, and VQA evidence identity. Identity parity, escalation logs, and grounded-answer/failure rates are the primary contract checks.

Offline replay is diagnostic, not a VBS leaderboard score, and uses a frozen manifest, split, labels, index/model revisions, hardware, configuration, and raw predictions. Known-item tasks report Recall@1/5/10/20, MRR, nDCG@10, target-video and frame/interval hit; AVS reports unique-video coverage and duplicate rate when annotations exist; VQA reports answer correctness, localization, grounded-answer rate, identity parity, and p50/p95 latency. Query, retrieval, reranking, VLM, end-to-end, and human-action latency remain separate. The three questions are whether evidence survives each handoff, whether escalation buys quality for its added time, and whether grounding reduces invalid exposure without hiding failures.

The ablation tests dense/payload/RRF, HyDE, reranking, temporal coherence, diversification, grounding, exact search, larger HNSW settings, and (when enabled) shot-level coarse-to-fine routing against the live default. Empty retrieval, decode failure, malformed response, timeout, provenance mismatch, and duplicate/wrong submission are counted with quality metrics. No result cell is populated until the manifest and index revision are frozen; this draft reports the protocol rather than invented numbers.

6 Limitations and Conclusion

The source-index migration leaves optional enrichment fields non-universal, and official-keyframe native frame mapping still needs an audit for frame-accurate submission. VLM verification may dominate p95 latency and provider revisions can change outputs. Ambient audio, secondary embeddings, shot routing, and verification are opt-in; the DRES adapter is not yet validated against the live instance; and the legacy CLI VQA route has a different guarantee. AVS and some KIS-C outcomes depend on operators and live judging. The release should publish the manifest, schema/version, configuration, model revisions, raw predictions, metric script, interaction trace, and failure logs.

TGLTW-RMIT’s novelty is not a new embedding, rank-fusion rule, or VLM. It is the explicit composition boundary: a stable evidence object crosses multimodal indexing, bounded task routing, operator-visible escalation, grounded VQA, and submission/logging. This makes live quality, latency, and failure modes measurable as properties of one system. The next milestone is frozen replay and on-site rehearsal without changing the live configuration.

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